229828: feldspathic sandstone, Killaloe prospect

(Mount Kirk Formation, Higginsville Subgroup, Kalgoorlie Group, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Widgiemooltha (SH 51-14), Cowan (3234)
MGA Zone 51, 395552E 6460086N

Sampled on 15 January 2019

This sample was collected from the 289.56 – 289.94 m depth interval of drillcore 14KLDH03, drilled in 2014 by Matsa Resources Limited at their Killaloe prospect (Breyley, 2015). The drillhole is located on the east side of Lake Cowan, about 16.5 km northwest of Buldania Rocks, 7.2 east-southeast of the abandoned Peninsula mine, and 2.6 km northeast of Killaloe Hill.

Geological context

The unit sampled is a feldspathic sandstone (Fig. 1), assigned to the Mount Kirk Formation of the 2753–2680 Ma Kalgoorlie Group (Hall et al., 2022). Within the Eundynie greenstone belt, the 2720–2690 Ma Mount Kirk Formation consists mainly of metabasalt, with lesser amounts of meta-ultramafic and metasedimentary rocks (Hayman et al., 2015; Sapkota, 2022). A mafic intrusion within the unconformably underlying Woolyeenyer Formation has been dated at 2714 ± 5 Ma (Hill et al., 1992). The present sample was collected from a graded bed of sandstone within a sequence of laminated carbonaceous shale (Breyley, 2015), with the aim of constraining the age of the succession of metamorphosed mafic–ultramafic and interflow sedimentary rocks.

Petrographic description

The sample is an altered feldspathic sandstone (Fig. 1), consisting of about 50–60% plagioclase, 30–40% K-feldspar, 7% lithic grains, 2% altered magnetite, 1% quartz and accessory zircon. Plagioclase is albitized and variably overprinted by carbonate minerals and sericite; microcline is strained. Lithic grains are subangular, up to 1 mm in size, and include microcrystalline cherty grains and some replaced by fine sericite. The framework grains are closely packed, there is no obvious matrix, and there is no indication of grading at the scale of the thin section. Feldspars are angular, not abraded, and up to 1.3 mm in size (average 0.6 mm). Quartz grains are subangular to rounded, up to 0.8 mm in diameter, and weakly strained. Altered magnetite is mainly anhedral, equant, and irregular, up to 0.4 mm in diameter, and altered to titanite and brown iron oxide minerals. Alteration consists of carbonate–sericite–albite–quartz. Where the alteration is most intense, only remnants of feldspar grains are evident in massive carbonate–sericite. Secondary quartz occurs as finely microcrystalline patches, probably after lithic grains. The abundance of plagioclase and microcline suggests a granitic provenance. The sandstone is altered but not obviously deformed or metamorphosed.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque, and anhedral to subhedral. The crystals are up to 400 µm long, and equant to elongate, with aspect ratios up to 4:1. Concentric zoning is ubiquitous, and many zircons contain inclusions of other minerals. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 24–25 September 2020, using SHRIMP-B. Twelve analyses of the M257 standard were obtained during the session, of which 10 indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.25% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Eighteen analyses were obtained from 18 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3), and form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 18 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2703 ± 6 Ma (MSWD = 1.9) and a discordia with intercepts at 2713 ± 8 and c. 700 Ma (MSWD = 1.2).

The date of 2713 ± 8 Ma for the 18 analyses in Group I is interpreted as the igneous crystallization age of the source rock that yielded the dominant detrital component and a maximum age of deposition for the sandstone.

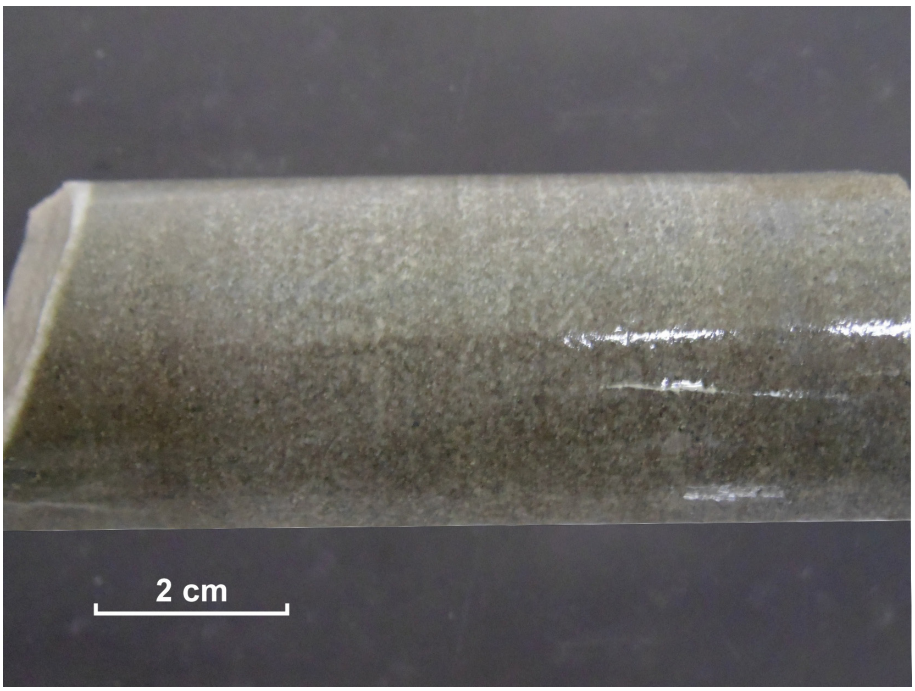

Figure 1. **Drillcore image (wet surface) of sample 229828: feldspathic sandstone, Killaloe prospect**

Figure 2. **Cathodoluminescence image of representative zircons from sample 229828: feldspathic sandstone, Killaloe prospect. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 229828: feldspathic sandstone, Killaloe prospect

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	10	10.1	217	69	0.32	0.365	2.090	0.020	0.18496	0.00097	2.098	0.020	0.18170	0.00152	2513	20	2669	14	5.8
I	16	16.1	257	189	0.74	0.141	2.011	0.018	0.18437	0.00083	2.014	0.018	0.18311	0.00089	2599	20	2681	8	3.1
I	8	8.1	245	69	0.28	0.054	1.936	0.017	0.18443	0.00081	1.937	0.017	0.18395	0.00084	2684	20	2689	8	0.2
I	3	3.1	192	41	0.22	0.074	2.111	0.036	0.18494	0.00097	2.113	0.036	0.18428	0.00101	2498	35	2692	9	7.2
I	5	5.1	188	48	0.25	-0.034	1.986	0.019	0.18446	0.00092	1.985	0.019	0.18476	0.00094	2630	21	2696	8	2.5
I	17	17.1	120	44	0.36	0.221	2.027	0.043	0.18686	0.00117	2.032	0.043	0.18489	0.00131	2580	46	2697	12	4.3
I	9	9.1	295	88	0.30	0.069	2.013	0.017	0.18559	0.00076	2.014	0.017	0.18497	0.00079	2599	19	2698	7	3.7
I	6	6.1	252	95	0.38	0.055	1.952	0.017	0.18569	0.00076	1.953	0.017	0.18520	0.00079	2665	20	2700	7	1.3
I	4	4.1	125	50	0.40	0.135	2.037	0.035	0.18663	0.00190	2.039	0.035	0.18542	0.00195	2572	36	2702	17	4.8
I	13	13.1	277	132	0.48	0.116	1.921	0.017	0.18659	0.00079	1.923	0.017	0.18556	0.00084	2699	20	2703	7	0.2
I	18	18.1	204	72	0.35	0.171	1.976	0.019	0.18761	0.00096	1.980	0.019	0.18609	0.00105	2636	22	2708	9	2.7
I	7	7.1	292	87	0.30	0.039	1.934	0.017	0.18676	0.00077	1.935	0.017	0.18641	0.00078	2685	19	2711	7	0.9
I	14	14.1	257	108	0.42	0.045	1.991	0.018	0.18681	0.00082	1.992	0.018	0.18641	0.00084	2623	19	2711	7	3.3
I	15	15.1	230	68	0.29	0.111	1.965	0.018	0.18746	0.00091	1.967	0.018	0.18647	0.00097	2649	21	2711	9	2.3
I	1	1.1	271	92	0.34	-0.086	1.949	0.017	0.18575	0.00076	1.947	0.017	0.18652	0.00080	2672	19	2712	7	1.5
I	12	12.1	209	69	0.33	0.033	1.960	0.019	0.18690	0.00141	1.961	0.019	0.18660	0.00142	2657	21	2713	13	2.1
I	2	2.1	101	29	0.29	0.483	1.965	0.022	0.19154	0.00127	1.975	0.022	0.18723	0.00155	2641	25	2718	14	2.8
I	11	11.1	242	75	0.31	0.009	1.893	0.017	0.18771	0.00085	1.893	0.017	0.18763	0.00086	2735	21	2722	8	-0.5

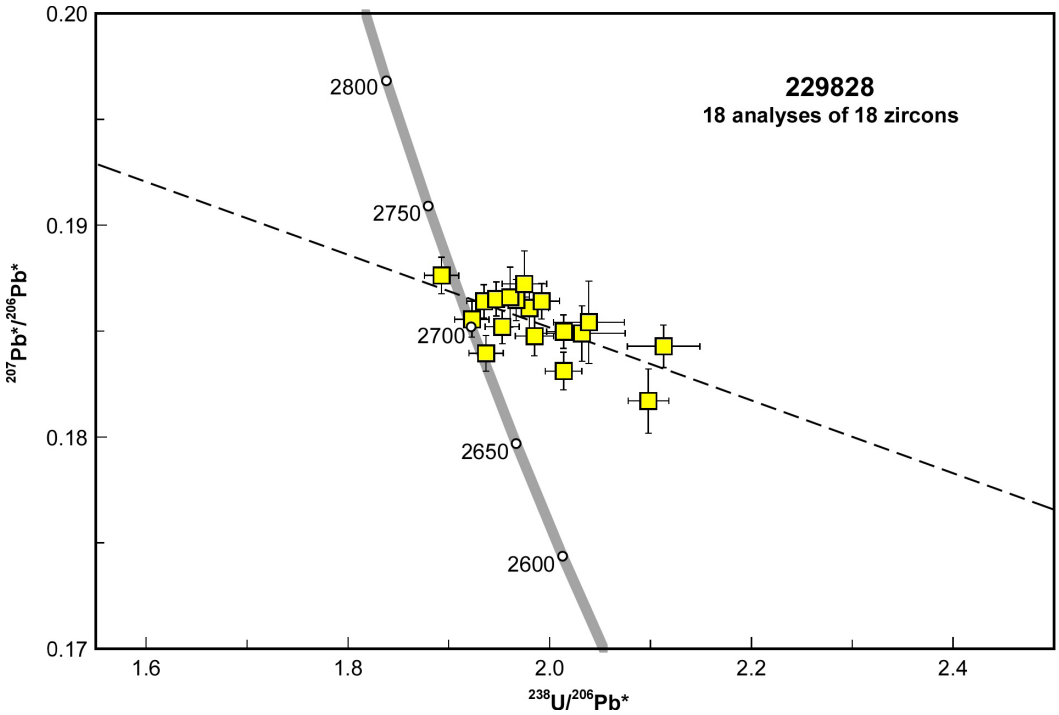

Figure 3. U–Pb analytical data for sample 229828: feldspathic sandstone, Killaloe prospect. Yellow squares indicate Group I (magmatic zircons). The dashed line indicates a discordia regression through data in Group I

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Recommended reference for this publication

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